SRI SRINIVASA EDUCATIONAL & CHARITABLE TRUST (R)
SAPTHAGIRI COLLEGE OF ENGINEERING
(Affiliated to Visvesvaraya Technological University, Belagavi and Approved by AICTE-New Delhi)
(Accredited by NAAC "A" Grade & NBA) (ISO 9001:2015 & ISO 14001:2015 Certified)
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Department of Mathematics
Assignment -I

Subject: Mathematics-I for ME/CV/EEE stream **Sub Code: BMATM/C/E101**

Q.NO	Assignment - I	BLT	CO's
1	Solve: $r\sin\theta - \cos\theta \frac{dr}{d\theta} = r^2$.	L1,L2&L3	CO3
2	Solve the equation $(px-y)(py+x) = 2p$ by reducing into Clairaut's form, taking the substitution $X = x^2, Y = y^2$.	L1,L2&L3	CO3
3	Find Orthogonal Trajectories of the family of curves $r^n = a^n \cos n\theta$.	L1,L2&L3	CO3
4	Test for consistency and solve: $x+y+z=6;\quad x-y+2z=5;\quad 3x+y+z=8$	L1,L2&L3	CO5
5	Solve the system of equations by Gauss-SeidelMethod: $20x+y-2z=17;\ \ 3x+20y-z=-18,\ 2x-3y+20z=25.$ Carry out 3 iterations taking the initial approximation to the solutions as (1,0,0).	L1,L2&L3	CO5
6	Using Rayleigh's power method find the largest Eigen value and the corresponding Eigen vector of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ with $X^{(0)} = [1 \quad 1 \quad 1]^T$ as the initial Eigenvector. (Perform 7 iterations)	L1,L2&L3	CO5